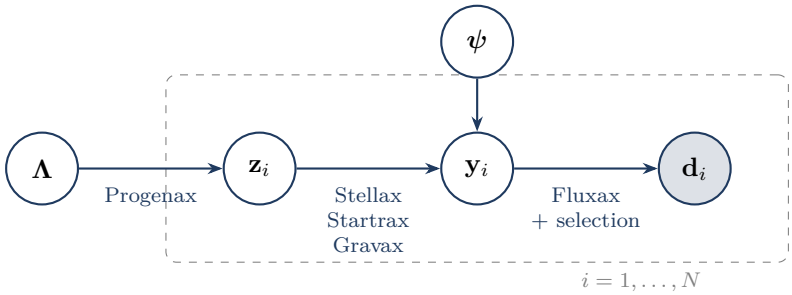
Lecture 7: Hierarchical models, selection effects, and checking inference

COMP 521 • Probability, Optimization, and Scientific Discovery • Sections 3.6–3.7

This handout is one part of the full COMP 521 notes. Section, equation, figure and listing numbers match the full notes, and references to anything outside this handout point there. The notation table is in "How to use these notes" (Lecture 1).

3.6 Hierarchical models for stellar populations

A *hierarchical* model separates what is shared from what is individual. Let $\boldsymbol{\Lambda}$ collect population hyperparameters (IMF slope, age spread, metallicity distribution, binary fraction), let $\mathbf{z}_i$ be star-specific latent variables (true initial mass, mass ratio of an unresolved companion), and let $\boldsymbol{\psi}$ be stellar or dynamical physics shared by every star.



Shaded nodes are observed; the dashed plate repeats for each star. If stars are conditionally independent given $(\boldsymbol{\Lambda}, \boldsymbol{\psi})$, the joint model factorizes as

$$p(\boldsymbol{\Lambda}, \boldsymbol{\psi}, \{\mathbf{z}_i\}, \{\mathbf{d}_i\}) = p(\boldsymbol{\Lambda})\, p(\boldsymbol{\psi}) \prod_{i=1}^{N} p(\mathbf{z}_i \mid \boldsymbol{\Lambda})\, p(\mathbf{d}_i \mid \mathbf{z}_i, \boldsymbol{\psi}). \tag{30}$$

Conditional independence is an assumption. Real clusters correlate their members through formation, dynamical evolution, mass segregation and binary interactions.

Missing physics is absorbed by the parameters you kept

If unresolved binaries are left out, the extra light they contribute is attributed to the IMF, and the fitted slope is biased. A hierarchical model with an explicit binary population makes the competition between explanations visible in the posterior, as a correlation between γ and the binary fraction.

3.6.1 Selection effects

Surveys see only stars that pass a detection criterion. Ignoring this is the most common error in population inference. Let $P_{\text{det}}(\mathbf{d})$ be the probability that a star with data $\mathbf{d}$ is detected and catalogued. The likelihood for one *detected* star is the population likelihood renormalized to the detectable part of the population [1, 2]:

$$p(\mathbf{d}_i \mid \boldsymbol{\Lambda}, \boldsymbol{\psi}, \text{det}) = \frac{P_{\text{det}}(\mathbf{d}_i)\, p(\mathbf{d}_i \mid \boldsymbol{\Lambda}, \boldsymbol{\psi})}{\alpha(\boldsymbol{\Lambda}, \boldsymbol{\psi})}, \qquad \alpha(\boldsymbol{\Lambda}, \boldsymbol{\psi}) = \int P_{\text{det}}(\mathbf{d})\, p(\mathbf{d} \mid \boldsymbol{\Lambda}, \boldsymbol{\psi})\, \mathrm{d}\mathbf{d}. \tag{31}$$

Here $p(\mathbf{d} \mid \boldsymbol{\Lambda}, \boldsymbol{\psi}) = \int p(\mathbf{d} \mid \mathbf{z}, \boldsymbol{\psi}) p(\mathbf{z} \mid \boldsymbol{\Lambda})\, \mathrm{d}\mathbf{z}$. The factor $P_{\mathrm{det}}(\mathbf{d}_i)$ does not depend on the parameters and drops out, but the *detection fraction* α does not: a steeper IMF puts more stars below the detection limit, and α is what tells the model so. α is usually computed by Monte Carlo, pushing simulated populations through the forward model and the survey's selection function. If the number of detected stars N_{obs} is itself informative, use the extended likelihood: multiply the product of Eq. (31) over detected stars by $(N_{\mathrm{exp}}\alpha)^{N_{\mathrm{obs}}} e^{-N_{\mathrm{exp}}\alpha}$, where N_{exp} is the expected total number of stars. The factor $\alpha^{N_{\mathrm{obs}}}$ cancels the per-star denominators, leaving $e^{-N_{\mathrm{exp}}\alpha}$ to carry the selection information.

3.7 Checking an inference code

A posterior can be precise and wrong. Two complementary checks catch different failures.

Posterior predictive checks [3] simulate replicated data from the posterior predictive and compare them with the observed data through a test statistic chosen to probe a suspected weakness (the colour spread on the giant branch, the number of stars above the turnoff). They test the *model*.

Simulation-based calibration (SBC) [4] tests the *inference code*, for example a sampler, an emulator-based likelihood, or a variational approximation. Repeat many times: draw $\tilde{\boldsymbol{\theta}}$ from the prior, simulate $\tilde{\mathbf{d}}$, run inference on $\tilde{\mathbf{d}}$ to get S posterior draws, and record the rank of $\tilde{\boldsymbol{\theta}}$ among them (the number of draws below it, an integer from 0 to S). If the inference is exact, the ranks are uniform, because averaging the posterior over data simulated from the prior returns the prior. Departures have recognizable shapes:

- U-shaped ranks: the posteriors are too narrow (overconfident);
- a central hump: too wide (underconfident);
- a slope: systematically biased.

ex07_sbc.py -- SBC for the conjugate model with three inference codes

```python
"""Simulation-based calibration (SBC) for the conjugate Gaussian model.

theta ~ N(0, 1),  y = 2 theta + eps,  eps ~ N(0, 1).
Exact posterior: N(2y/5, 1/5).  We test three 'inference codes':
  correct, overconfident (sd halved), and biased (mean shifted by +0.3 posterior sd).
If an inference code is calibrated, the rank of the true theta among S posterior
draws is uniform on {0, ..., S}.
"""
import jax
import jax.numpy as jnp

jax.config.update("jax_enable_x64", True)

n_sims, S = 4000, 99


def posterior(y, kind):
    mean, sd = 2.0 * y / 5.0, jnp.sqrt(0.2)
    if kind == "overconfident":
        sd = 0.5 * sd
    if kind == "biased":
        mean = mean + 0.3 * sd
```

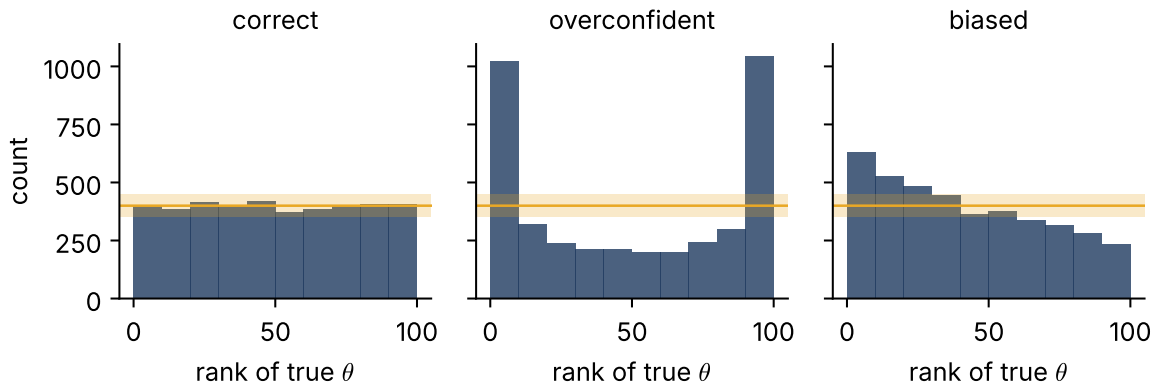


Figure 6: SBC rank histograms (4000 simulations, 99 posterior draws each). The gold band is the 99% range for a bin count under uniformity.

```
    return mean, sd


def one_sim(key, kind):
    k1, k2, k3 = jax.random.split(key, 3)
    theta = jax.random.normal(k1)  # draw from the PRIOR
    y = 2.0 * theta + jax.random.normal(k2)  # simulate data
    mean, sd = posterior(y, kind)
    draws = mean + sd * jax.random.normal(k3, (S,))  # 'run inference'
    rank = jnp.sum(draws < theta)
    covered = jnp.abs(theta - mean) < sd  # nominal 68.3% central interval
    return rank, covered


keys = jax.random.split(jax.random.key(11), n_sims)
results = {}
for kind in ["correct", "overconfident", "biased"]:
    ranks, cov = jax.vmap(lambda k: one_sim(k, kind))(keys)
    counts = jnp.histogram(ranks, bins=10, range=(0, S + 1))[0]
    chi2 = jnp.sum((counts - n_sims / 10) ** 2 / (n_sims / 10))
    results[kind] = ranks
    print(f"{kind:>13}: 68% interval coverage = {cov.mean():.3f};"
          f" rank-histogram chi^2 = {chi2:6.1f} (9 dof; 95th pct = 16.9)")
```

Output (executed for these notes):

```
      correct: 68% interval coverage = 0.674; rank-histogram chi^2 =    4.8 (9 dof; 95th pct = 16.9)
 overconfident: 68% interval coverage = 0.380; rank-histogram chi^2 = 2553.5 (9 dof; 95th pct = 16.9)
       biased: 68% interval coverage = 0.664; rank-histogram chi^2 =  333.9 (9 dof; 95th pct = 16.9)
```

The biased code is the instructive one. Its 68% intervals cover the truth 66% of the time, close to nominal, yet its rank histogram fails decisively. Coverage at one level is a weak test; the full rank distribution is a strong one. SBC tests calibration averaged over the prior. It does not test whether the model describes nature; that is the job of posterior predictive checks and independent data.

Exercise for this lecture

Exercise 13, numbered as in Appendix A of the full notes. Worked solutions are in Appendix B; attempt each one before reading its solution.

13. [C] SBC finds what coverage misses. Apply SBC to the Laplace approximation of a Poisson log-rate: $\theta \sim \mathcal{N}(0, 1)$, $n \sim \text{Poisson}(Te^{\theta})$ with $T = 3$. Report the rank histogram, its χ^2, and 68% coverage. Interpret the shape of the histogram in terms of the true posterior when $n = 0$.

References

[1] T. J. Loredo. Accounting for source uncertainties in analyses of astronomical survey data. In *Bayesian Inference and Maximum Entropy Methods in Science and Engineering*, volume 735 of *AIP Conference Proceedings*, pages 195–206, 2004.

[2] I. Mandel, W. M. Farr, and J. R. Gair. Extracting distribution parameters from multiple uncertain observations with selection biases. *Monthly Notices of the Royal Astronomical Society*, 486:1086–1093, 2019.

[3] A. Gelman, X.-L. Meng, and H. Stern. Posterior predictive assessment of model fitness via realized discrepancies. *Statistica Sinica*, 6:733–807, 1996.

[4] S. Talts, M. Betancourt, D. Simpson, A. Vehtari, and A. Gelman. Validating Bayesian inference algorithms with simulation-based calibration. arXiv:1804.06788, 2018.